

7.7 HIGH-PERFORMANCE BUILDINGS AND SITE RESILIENCE

Objectives

- (a) To ensure a high level of sustainability across all elements of the estate.
- (b) To optimize opportunities for environmentally sustainable design to minimise carbon emissions, energy use, potable water use and waste.
- (c) To encourage a low-carbon, high-performance precinct to help Waverley work towards its target of net-zero emissions.
- (d) To ensure buildings are well-designed to minimise energy consumption as well as maximise the thermal comfort for the occupants.
- (e) To promote the use of on-site energy generation and storage.
- (f) To ensure the site and buildings are designed to reduce potable water consumption.
- (g) To protect water quality and promote appropriate water harvesting and on-site storage and use of harvested/recycled water.
- (h) To minimise the impacts of the urban heat island effect.
- (i) To minimise the impacts of drought or water shortages.
- (j) To minimise impacts from severe storms or flooding events.
- (k) To support key environmental targets in Council's strategic plans.

Controls

7.7.1 Energy use and production

- (a) Building location and façade design is to maximise access to direct sunlight to reduce reliance on mechanical heating.
- (b) Adequate external shading or performance glass is to be provided on the western and north-western building facades to minimise the cooling load required in mid-summer.
- (c) Any on-site renewable energy sources are to be coupled with battery storage.
- (d) The residential component of a building is expected to exceed BASIX Energy, Water and Thermal Comfort targets as specified in the WLEP.
- (e) Commercial buildings shall achieve a Green Star Certified Rating of six (6) stars; and/or a NABERS 5.5-star energy rating and 4.5-star water rating as specified in the WLEP.

7.7.2 Urban Heat Island

- (a) To mitigate the accumulation of urban heat, buildings are to utilise light coloured and reflective materials, and where possible provide vegetative surfaces such as green roofs.
- (b) Buildings and roof materials are to be a light coloured material to reduce solar absorption.
- (c) Pavements are to mitigate heat accumulation where possible through materials which consider end users and that maximize permeability and/or reflectivity.
- (d) Provide one building in the precinct able to function as an accessible refuge for on-site residents from the heat in the case of emergency, with access to a back-up power supply available on the site.